

MAIN 主要

HARDWARE 硬件

SIM : Subscriber Identity Module 用户识别卡
IMSI : International Mobile Subscriber Identity 国际移动用户识别码
IMEI : International Mobile Equipment Identity 国际移动设备识别码
MEID : Mobile Equipment Identifier 移动设备识别码 用于CDMA制式的手机
ICCID : Integrated circuit card identity 集成电路卡识别码

NETWORK 网络

MCC : Mobile Country Code 移动国家码
MNC : Mobile Network Code 移动网络号码
5G SA : StandAlone 独立组网 NR
5G NSA : Non-StandAlone 非独立组网 NR+LTE
VoLTE : Voice over LTE 高清通话/长期演进语音承载
VoNR : Voice over New Radio 新空口承载语音
CSFB : Circuit Switch Fallback 电路域回落
TDD : Time Division Duplexing 时分双工
FDD : Frequency Division Duplexing 频分双工
NB-IOT : NarrowBand Internet of Things 窄带物联网
UMTS : Universal Mobile Telecommunications System 通用移动通信系统(TDSCDMA|WCDMA)
EVDO : Evolution Data Only CDMA1xEVDO 3G 发展—仅数据

CELL TYPE 小区类型

NR : New Radio 新空口
LTE : Long Term Evolution 长期演进技术
WCDMA : Wideband Code Division Multiple Access 宽带码分多址
TDSCDMA : Time Division-Synchronous Code Division Multiple Access 时分同步的码分多址
CDMA : Code Division Multiple Access 码分多址
GSM : Global System for Mobile Communications 全球移动通信系统

- ⇒ CELL 小区
- ⇒ .JSON
- ⇒ .CSV

CELL 小区

Type 小区类型

```
NR | LTE | GSM | WCDMA | TDSCDMA | CDMA | NONE
```

NR

```
int nr_tac // 0..65535 16-bit Tracking Area Code 追踪区域码
int nr_pci // 0..1007 Physical Cell Identifier 物理小区标识
long nci // 0..68719476735 36-bit Cell Identifier 小区标识
int nrafcn // 0..3279165 Absolute Radio Frequency Channel Number 绝对无线频道编号
// Channel state information 信道状态信息
int csi_rsrp // -140..-44 dBm Reference Signal Received Power 参考信号接收功率
int csi_rsrq // -20..-3 dB Reference Signal Receiving Quality 参考信号接收质量
int csi_sinr // -23 ..23 dB Signal to Interference plus Noise Ratio 信号与干扰加噪比
// Synchronization Signal 同步信号
int ss_rsrp // -140 ..-44 dBm Reference Signal Received Power 参考信号接收功率
int ss_rsrq // -20 ..-3 dB Reference Signal Receiving Quality 参考信号接收质量
int ss_sinr // -23 ..40 dB Signal to Interference plus Noise Ratio 信号与干扰加噪比
```

LTE

```
int tac // 16-bit Tracking Area Code
int pci // 0..503 Physical Cell Identifier 物理小区标识
int eci // 28-bit E-UTRAN Cell Identifier 小区全局标识符 ECI (28 bits) = eNB ID
int earfcn // 18-bit E-UTRA Absolute Radio Frequency Channel Number 载波频点号
int band_width // kHz Android 9.0+ 频宽

int rssi // -113..-51 Received Signal Strength Indicator 接收信号强度指示
int rsrp // -140..-43 dBm Reference Signal Received Power 参考信号接收功率
int rsrq // -20..-3 dB Reference Signal Receiving Quality 参考信号接收质量
float sinr // -20..30 dB Signal to Interference plus Noise Ratio 信号与干扰加噪比
int cqi // 0..15 Channel Quality Indicator 信道质量指示
```

GSM | TDSCDMA | WCDMA

```
int lac // 0..65535 16-bit Location Area Code 位置区码
int ci // (0..65535 16-bit GSM) (0..268435455 28-bit UMTS) Cell Identity 小区标识
int rxlev // -113..-51 dBm Received Signal Level 接收信号电平
int arfcn // 16-bit GSM Absolute Radio Frequency Channel Number 绝对无线频道编号
int bsic // 6-bit Base Station Identity Code 基站识别码
int psc // 0..511 9-bit UMTS Primary Scrambling Code 主扰码
int uarfcn // 16-bit UMTS Absolute Radio Frequency Channel Number 绝对无线频道编号
```

CDMA

```
int    nid // 0..65535 Network Id 网络识别码
int    bid // 0..65535 Base Station Id 基站识别码
int    sid // 0..32767 System Id 系统识别码
int    cdma dbm // -120..0 RSSI Received Signal Strength Indicator 接收信号的强度指示
float  cdma ecio // -160..0 Ec/Io EnergyChip/InterfereOther 接收信号的强度和邻小区干扰水
int    evdo dbm // -120..0 RSSI Received Signal Strength Indicator 接收信号的强度指示
float  evdo ecio // -160..0 Ec/Io EnergyChip/InterfereOther 接收信号的强度和邻小区干扰水
int    evdo snr // 0..8 signal to noise ratio 信号与噪声比
```

⇒ MAIN 主要

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. CSV SAMPLE 示例

POST Headers

```
Content-Type:text/plain
X-Client-Type:Android
VERSION-NAME:6.x.x
CELL-VIP-ID:xxxxxxx
SLOT-INDEX:1 (1 OR 2)
OPERATOR:xxxxx (MCC+MNC 46000\46001...)
UUID:xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxx (UNIQUE FILE ID 唯一文件ID)
```

Body raw

```
"TIME","LATITUDE","LONGITUDE","TYPE","NR-TAC","NR-PCI","NR-CI","NR-ARFCN","CSI-RS
"20220721 10:19:00.516","40.000000","115.000000","NR","2000000","690","574444444"
"20220721 10:19:00.640","40.000000","115.000000","NR","2000000","690","574444444"
"20220721 10:19:00.649","40.000000","115.000000","NR","2000000","690","574444444"
"20220721 10:19:00.697","40.000000","115.000000","NR","2000000","690","574444444"
"20220721 10:19:00.700","40.000000","115.000000","NR","2000000","690","574444444"
```

Body = TIME + LOCATION + CELL

LOCATION

```
LATITUDE float
LONGITUDE float
```

CELL小区 COLUMNS

```
TYPE,
NR-TAC,NR-PCI,NR-CI,NR-ARFCN,
CSI-RSRP,CSI-RSRQ,CSI-SINR,SS-RSRP,SS-RSRQ,SS-SINR,
TAC,PCI,ECI,EARFCN,BAND_WIDTH,RSSI,RSRP,RSRQ,SINR,CQI,
LAC,CI,RXLEV,ARFCN,BSIC,PSC,UARFCN,
NID,BID,SID,CDMADB,CDMAECIO,EVDODBM,EVDOECIO,EVDOSNR
```

- ⇒ CELL 小区
- ⇒ MAIN 主要
- ⇒ .JSON

.JSON SAMPLE 示例

POST Headers

```
Content-Type:application/json
X-Client-Type:Android
VERSION-NAME:6.x.x
CELL-VIP-ID:xxxxxxx
SLOT-INDEX:1 (1 OR 2)
OPERATOR:xxxxx (MCC+MNC 46000\46001...)
UUID:xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxx (UNIQUE FILE ID 唯一文件ID)
```

Body raw

```
{
  "cell": {
    "type": "NR",
    "nci": xxxxxxxxxx,
    "nr_pci": xxx,
    "nr_tac": xxxxxxx,
    "nrarfcn": 504990,
    "ss_rsrp": -66,
    "ss_rsrq": -11,
    "ss_sinr": 26
  },
  "datanet": "NR",
  "is_data_slot": 1,
  "is_voice_slot": 0,
  "mcc": "460",
  "mnc": "00",
  "ncells": [
    {
      "type": "NR",
      "nr_tac": xxxxxxx,
      "ss_rsrp": -66
    },{
      "type": "NR",
      "nr_tac": xxxxxxx,
      "ss_rsrp": -66
    }
  ],
  "netoperator": "XXXX",
  "simoperator": "XXXX",
  "slot_sn": 0,
  "voicenet": "UNKNOWN",
  "time": "20220721 10:13:04",
  "longitude": "xxx.xxxxxx",
  "latitude": "xx.xxxxxx",
```

```
"address": "XXXXXXXXXXXXXXXXXXXX"
```

```
}
```

Body Parts

```
cell : Serving Cell 主服务小区  
ncells : Neighbor Cells 邻小区列表  
slot_sn : 0 OR 1
```

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